Fiscal Year:	FY 2017	**Task Last Updated:**	FY 11/30/2017
PI Name:	Sabanayagam, Chandran Ph.D.		
Project Title:	Epigenetic and Protein Expression Pattern Profiling of Caenorhabditis elegans Exposed to Time-Varying Gravitational Fields: a Multi-Generational Study		
Division Name:	Space Biology		
Program/Discipline:	SPACE BIOLOGY		
Program/Discipline--Element/Subdiscipline:	SPACE BIOLOGY--Developmental biology		
Joint Agency Name:		**TechPort:**	No
Human Research Program Elements:	None		
Human Research Program Risks:	None		
Space Biology Element:	(1) Cell & Molecular Biology (2) Animal Biology: Invertebrate		
Space Biology Cross-Element Discipline:	(1) Reproductive Biology (2) Developmental Biology (3) Musculoskeletal Biology (4) Neurobiology		
Space Biology Special Category:	None		
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Zip Code:	19711-5449	**Congressional District:**	1
Comments:			
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No. of Master's Candidates:	1	**No. of Bachelor's Degrees:**	0
No. of Bachelor's Candidates:	0	**Monitoring Center:**	NASA ARC
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Flight Program:			
Flight Assignment:	NOTE: Extended to 8/31/2017 per NSSC information (Ed., 9/1/16)		
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COI Name (Institution):			
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Task Description:	We hypothesize that long term temporal changes in the gravity vector will result in animal adaptation via epigenetic mechanisms that are passed on to subsequent generations. We further hypothesize that the gravity-influenced epigenetic profile can be restored to the consensus wild-type (i.e., Earth born) upon introduction to stable 1xg conditions. We propose to utilize C. elegans as a model organism to address this hypothesis by performing the following experiments: 1) Determine C. elegans behavioral responses (i.e., changes in migration dynamics, animal body posture, metabolism, etc.) due to temporal changes in the gravity vector. The gravity vector will be controlled by placing the animals inside a rotating CubeSat microscope developed by our group. The frequency, direction, and duration of rotation will be varied to elicit the most dramatic behavioral responses. Habituation time will be evaluated, and also changes in lifespan. 2) Identify gravity responsive genes by utilizing the vast collection of fluorescent strains of C. elegans available to researchers. We will monitor gene expression by fluorescence microscopy--from the embryonic stage through adulthood--during changes in the gravity vector. Genes of interest include those associated with muscle development, cell apoptosis, and neuronal development. 3) Profile the epigenetic marks on C. elegans exposed to long periods of continuous gravity vector changes that range from 1/4 of an animal's lifespan to three generations. We will examine the epigenetic consequence of restoring constant 1xg conditions over successive generations. The three Specific Aims can be conducted with the same animal strains, and thus allows the opportunity to correlate the findings across different experiments.
Rationale for HRP Directed Research:	
Research Impact/Earth Benefits:	Understanding gravity's effect on biological systems is not well studied because this parameter is difficult to manipulate during experiments. The International Space Station houses the newest U.S. national laboratory, which can be utilized to understand the influence of microgravity on molecular, cellular, and animal model systems. Determining changes in epigenetic profiles during multi-generations exposed to different environments is important to know to prepare for future manned spaceflight beyond Earth. Additionally, temporary alteration of epigenomes is relevant to human health, and similar adaptation mechanisms may be displayed by almost any change in the environment or diet. In order to expedite molecular and cellular research in microgravity, we developed an innovative clinorotation microscope that enables us to observe living organisms, in real-time, during microgravity. Because of the extended clinorotation times (4 days to 3 months), we also developed a novel automated "worm chip" that permits hands-free feeding and maintenance of the animals, without interrupting clinorotation.
Task Progress:	Caenorhabditis elegans is a unique model organism for Space Biology because it can live and reproduce in terrestrial and aquatic environments, enabling us to study how biology reacts to altered gravity using a rotating liquid-filled vessel (clinostat) in ground-based labs. Only a few model organisms exist that can be used in clinorotation experiments, but they can provide substantial insight into understanding how the gravitational vector affects fundamental biological processes. Many of the differentially regulated genes in clinorotated animals are expected to show similar behavior in true microgravity environments, and thus clinorotation can aid in the rational design of a set of biomarkers that will ultimately be tested in the space environment. Our group recently coined the term “gravitome” to represent the network of gravity-responsive genes. Understanding the components of the gravitome and how they function will be essential in developing technologies to support human health during space exploration. This study leverages the gene transcription data provided in GeneLab to compare the different spaceflight experiments with C. elegans onboard the International Space Station and our clinorotation experiments. We believe that this data is the first that presents comparative genomics across multiple GeneLab datasets. A lab-built clinostat was used for simulated microgravity experiments. Animals were synchronized by bleaching and ~10,000 embryos were cultured in VueLife culture bags (Standard C Series, Saint Gobain) containing ~50 mL of CeHR medium. Clinorotation was performed at a speed of 15 revolutions per minute for 10 days. After 10 days, animals were harvested for DNA, or maintained for up to three more generations (15 days) without clinorotation to simulate return to ground conditions. This project provided insight to the major Decadal Identifier AH16: Studies on transmission across generations of structural and functional changes induced by exposure to space during development. Our initial hypothesis that exposure to simulated microgravity will change gene expression levels, and that these changes will persist through multiple generations was confirmed primarily by RNA sequencing. We find that C. elegans embryos raised during clinorotation (F1 generation) elicit the transcription of new genes that were are not expressed in the parent (P0, ground control), and many of these genes are non-coding (nc)RNAs with unknown function. These new genes are still expressed in generations F2 through F4 raised in ground conditions, demonstrating a powerful epigenetic memory of the microgravity environment. It is important to note that these ncRNAs are not typically included in “standard” gene hybridization arrays (e.g., Affymetrix), but are identified using next-generation sequencing (NGS) methods. Future flight experiments should favor NGS over hybridization arrays because this will enable a complete representation of the transcriptome. Our RNA expression profiles show that upon return to ground conditions, about 1500 genes are differentially expressed for two generations; by the third generation on ground this number drops sharply to 104 genes. Thus, for the case of C. elegans and clinoroation, the exposure to altered gravity persists for about three generations. We compared our simulated microgravity results with similar flight experiments involving C. elegans in the NASA GeneLab database (four flight experiments + our clinorotation). The largest differentially expressed gene overlap between all the experiments was with the CERISE 4 spaceflight experiments and our clinorotation data. These results support the use of clinorotation as a ground-based analog that captures some aspects of true space microgravity, quantitated through gene expression profiling. It might be expected that CERISE 4 and CERISE 8 should be the most similar, but the experiments differed in that 9,000 animals were used in CERISE 4, while only 30-50 were used for CERISE 8. A low number of animals will produce highly variable DNA microarray results, which may have been the case for the CERISE experiments. This project provided insight to additional biological identifiers listed below. CMM-1a: How do alterations in gravity affect cell and microbial replication including genetic and metabolic regulation? Cell replication and reproduction of the microscopic worm (C. elegans) did not appear to be significantly altered in simulated microgravity, as determined by transmission light microscopy. OCB-1: What are the organs and systems that life uses to sense and react to gravity? Our fluorescence microscopy results show that many different tissues show differentially expressed genes after exposure to microgravity. The myo-3 gene

	was observed to be down-regulated based on fluorescence intensity levels compared to ground control. Also, abnormalities in vulva organ development was identified using pacr-2, a gene involved with fatty acid oxidation. OCB-2, CMM-5: Do changes in the gravity affect the basic metabolic rate and metabolism of living systems? What are the genetic changes underlying physiological responses to spaceflight demonstrated in biological systems? Our results indicate that metabolism is affected by clinorotation. For example, pcca-1 is a gene involved in propanoyl-CoA degradation in mitochondria and is down-regulated by clinorotation. We also find that the longevity pathway is down-regulated in simulated microgravity, leading to an increase in animal longevity. We hypothesize that the predicted increase in animal longevity is correlated with a decrease in sphingolipid production, based on bioinformatics analysis. OCB-5: What are the genetic changes associated with responses to spaceflight? Clinorotation produces genetic changes in the form of differential gene transcription. Interestingly, new genes not previously expressed by the parent are transcribed upon exposure to simulated microgravity. We present a new finding that many of these gene are non-coding RNA molecules, with unknown functions, as of yet. Also, changes in the chemical histone-tail modifications show evidence for regulating genes. We have used five histone chemical modifications to understand their dynamics under different environmental conditions, but there is no clear correlation between specific histone modifications and gene transcription. We found that exposure to different environments causes numerous histone modifications in the promoter regions of many genes, and this multivalency may serve as a “histone code” that regulates gene transcription over multiple generations. Because we probed only five different chemical modifications, out of billions of potential combinations, we have not been able to identify meaningful correlations between histone modifications and gene transcription. DEV-2: Can complex organisms be grown successfully through at least two complete life cycles in altered gravity environments? If not, why not? We have grown C. elegans for a month under clinorotation (~7 generations) and have observed normal growth population dynamics. These animals, however, need to adapt genetically to the new environment. Several articles have been submitted for publication as follows: Celen, I., Doh, J.H. & Sabanayagam, C.R. Genetic Adaptation of C. elegans to Environment Changes: Multigenerational Analysis of the Transcriptome (in review, Scientific Reports). Celen, I., Doh, J.H. & Sabanayagam, C.R. Multigenerational Genome-wide Chromatin Remodeling in C. elegans Under Environmental Changes (in review, Scientific Reports). Moore, A.B., Doh, J.H & Sabanayagam, C.R. The Big Brother Assay: high temporal imaging of individual C. elegans (submitted to BMC Biotechnology) Celen, I., Doh, J.H., Jayasinghe, A., Moore, M.T. & Sabanayagam, C.R. A Comparative Study of Genome-Wide responses to Microgravity (submitted to Nature Microgravity).
Bibliography Type:	Description: (Last Updated: 08/17/2023)
Articles in Peer-reviewed Journals	Çelen I, Doh JH, Sabanayagam CR. "Effects of liquid cultivation on gene expression and phenotype of C. elegans." BMC Genomics. 2018 Jul 31;19(1):562. https://doi.org/10.1186/s12864-018-4948-7 ; PubMed PMID: 30064382; PubMed Central PMCID: PMC6069985 , Jul-2018
Articles in Peer-reviewed Journals	Çelen I, Jayasinghe A, Doh JH, Sabanayagam CR. "Transcriptomic signature of the simulated microgravity response in Caenorhabditis elegans and comparison to spaceflight experiments." Cells. 2023 Jan 10;12(2):270. https://doi.org/10.3390/cells12020270 ; PMID: 36672205; PMCID: PMC9856674 , Jan-2023